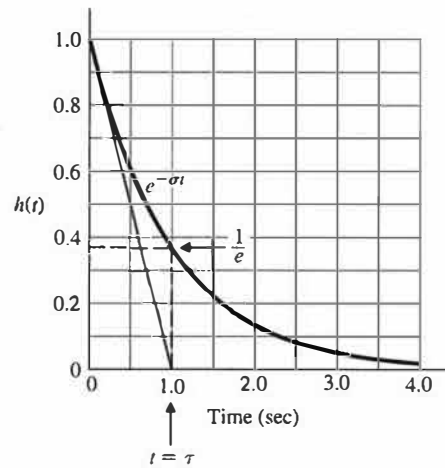


Figure 2.2
First-order system
response



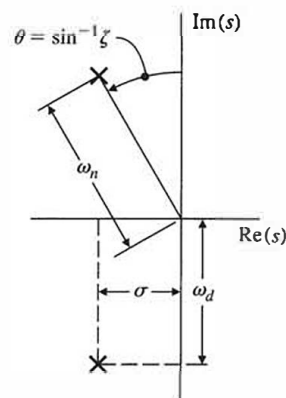
By expanding the form given by Eq. (2.12) and comparing with the coefficients of the denominator of $H(s)$ in Eq. (2.13), we find the correspondence between the parameters to be

$$\sigma = \zeta \omega_n \quad \text{and} \quad \omega_d = \omega_n \sqrt{1 - \zeta^2}, \quad (2.14)$$

damping ratio

where the parameter ζ is called the **damping ratio**, and ω_n is called the **undamped natural frequency**. The poles of this transfer function are located at a radius ω_n in the s -plane and at an angle $\theta = \sin^{-1} \zeta$, as shown in Fig. 2.3. Therefore, the damping ratio reflects the level of damping as a fraction of the critical damping value where the poles become real. In rectangular coordinates, the poles are at $s = -\sigma \pm j\omega_d$. When $\zeta = 0$ we have no damping, $\theta = 0$, and ω_d , the damped natural frequency, equals ω_n , the undamped natural frequency.

Figure 2.3
 s -plane plot for a pair of
complex poles



For the purpose of finding the time response corresponding to a complex transfer function from Table B.2, it is easiest to manipulate the $H(s)$ so that the complex poles fit the form of Eq. (2.12), because then the time response can be found directly from the table. The $H(s)$ from Eq. (2.13) can be written as

$$H(s) = \frac{\omega_n^2}{(s + \zeta\omega_n)^2 + \omega_n^2(1 - \zeta^2)},$$

therefore, from Entry 21 in Table B.2 and the definitions in Eq. (2.14), we see that the impulse response is

$$h(t) = \omega_n e^{-\sigma t} \sin(\omega_d t) 1(t).$$

For $\omega_n = 3$ rad/sec and $\zeta = 0.2$, the impulse response time history could be obtained and plotted by the MATLAB statements:

```
Wn = 3;
Ze = 0.2;
s=tf('s');
sys = (Wn^2)/(s^2+2*Ze*Wn*s+Wn^2)
impz(sys);
```

step response

It is also interesting to examine the **step response** of $H(s)$, that is, the response of the system $H(s)$ to a unit step input $u = 1(t)$ where $U(s) = \frac{1}{s}$. The step response transform given by $Y(s) = H(s)U(s)$, contained in the tables in Entry 22, is

$$y(t) = 1 - e^{-\sigma t} \left(\cos \omega_d t + \frac{\sigma}{\omega_d} \sin \omega_d t \right), \quad (2.15)$$

where $\omega_d = \omega_n \sqrt{1 - \zeta^2}$ and $\sigma = \zeta \omega_n$. This could also be obtained by modifying the last line in the MATLAB description above for the impulse response to

```
step(sys)
```

Figure 2.4 is a plot of $y(t)$ for several values of ζ plotted with time normalized to the undamped natural frequency ω_n . Note that the actual frequency, ω_d , decreases slightly as the damping ratio increases. Also note that for very low damping the response is oscillatory, while for large damping (ζ near 1) the response shows no oscillation. A few step responses are sketched in Fig. 2.5 to show the effect of pole locations in the s -plane on the step responses. It is very useful for control designers to have the mental image of Fig. 2.5 committed to memory so that there is an instant understanding of how changes in pole locations influence the time response. The negative real part of the pole, σ , determines the decay rate of an exponential envelope that multiplies the sinusoid. Note that if σ

Figure 2.4
Step responses of
second-order systems
versus ζ

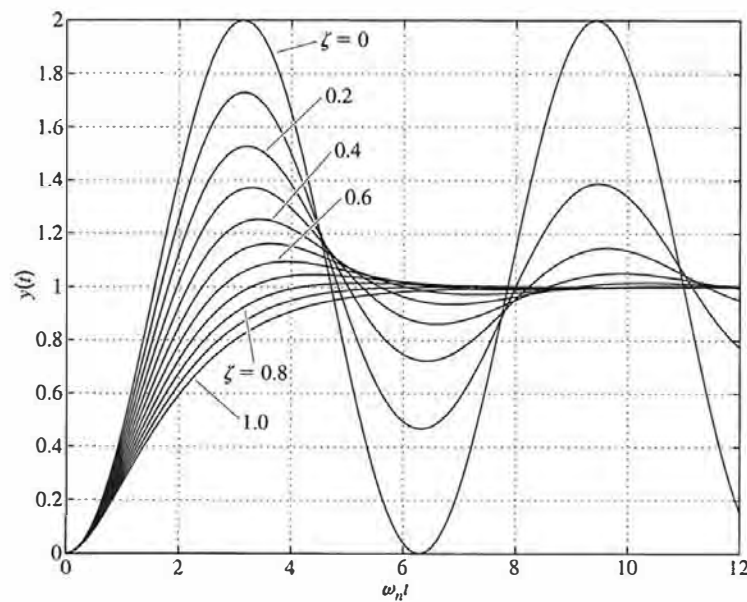
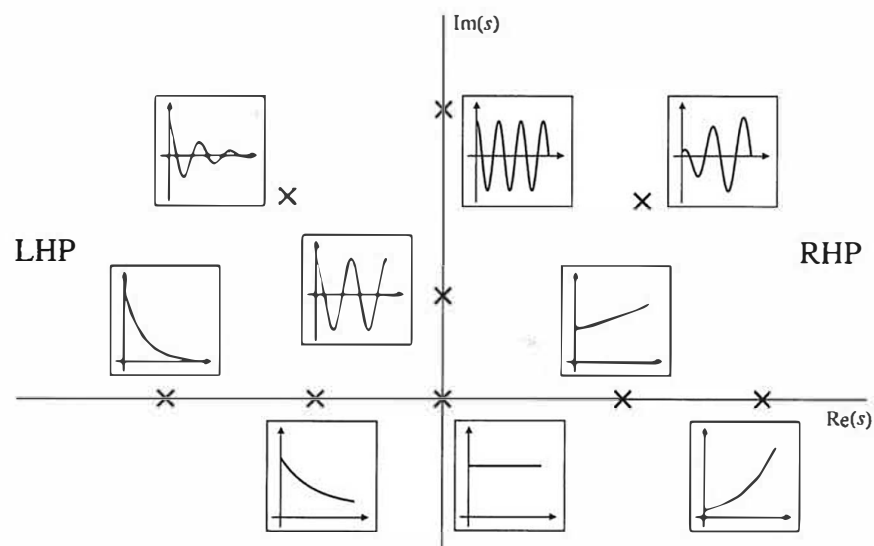


Figure 2.5
Time functions
associated with points in
the s -plane



is negative, the pole is in the right-half plane, the response will grow with time, and the system is said to be unstable. If $\sigma = 0$, the response neither grows nor decays, so stability is a matter of definition. If σ is positive, the natural response

decays and the system is said to be stable. Note that, as long as the damping is strictly positive, the system will eventually converge to the commanded value.

All these notions about the correspondence between pole locations and the time response pertained to the case of the step response of the system of Eq. (2.13), that is, a second-order system with no zeros. If there had been a zero, the effect would generally be an increased overshoot; the presence of an additional pole would generally cause the response to be slower. If there had been a zero in the right-half plane, the overshoot would be repressed and the response would likely go initially in the opposite direction to its final value. Nevertheless, the second-order system response is useful in guiding the designer during the iterations toward the final design, no matter how complex the system is.

2.1.7 Time-Domain Specifications

Specifications for a control system design often involve certain requirements associated with the time response of the system. The requirements for a step response are expressed in terms of the standard quantities illustrated in Fig. 2.6:

 t_r

The **rise time** t_r is the time it takes the system to reach the vicinity of its new set point.

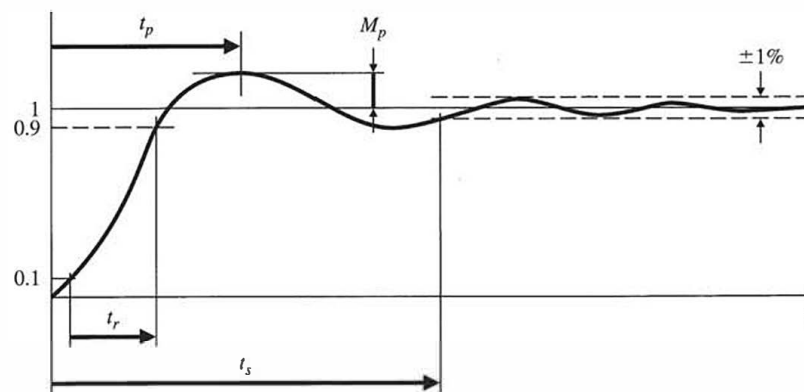
 t_s

The **settling time** t_s is the time it takes the system transients to decay.

 M_p

The **overshoot** M_p is the maximum amount that the system overshoots its final value divided by its final value (and often expressed as a percentage).

Figure 2.6
Definition of rise time t_r ,
settling time t_s , and
overshoot M_p



For a second-order system, the time responses of Fig. 2.4 yield information about the specifications that is too complex to be remembered unless approximated. The commonly used approximations for the second-order case with no zeros are

$$t_r \approx \frac{1.8}{\omega_n} \quad (2.16)$$

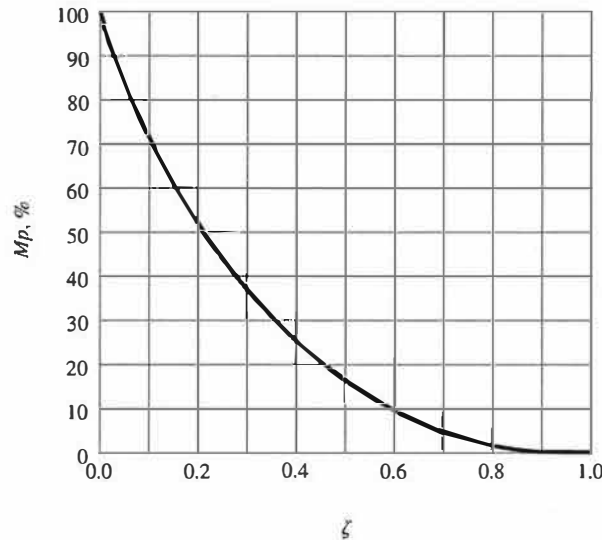
$$t_s \approx \frac{4.6}{\zeta \omega_n} = \frac{4.6}{\sigma} \quad (2.17)$$

$$M_p \approx e^{-\pi\zeta/\sqrt{1-\zeta^2}} \quad 0 \leq \zeta < 1 \quad (2.18)$$

The overshoot M_p is plotted in Fig. 2.7. Two frequently used values from this curve are $M_p = 16\%$ for $\zeta = 0.5$ and $M_p = 5\%$ for $\zeta = 0.7$.

Equations (2.16)–(2.18) characterize the transient response of a system having no finite zeros and two complex poles with undamped natural frequency ω_n , damping ratio ζ , and negative real part σ . In analysis and design, they are used to obtain a rough estimate of rise time, overshoot, and settling time for just about any system. It is important to keep in mind, however, that they are qualitative guides and not precise design formulas. They are meant to provide a starting point for the design iteration and the time response should always be checked after the control design is complete by an exact calculation, usually by numerical simulation, to verify whether the time specifications are actually met. If they have not been met, another iteration of the design is required. For example, if the rise

Figure 2.7
Plot of the peak overshoot M_p versus the damping ratio ζ for the second-order system



time of the system turns out to be longer than the specification, the target natural frequency would be increased and the design repeated.

2.2 Basic Properties of Feedback

An open-loop system described by the transfer function $G(s)$ can be improved by the addition of feedback including the dynamic compensation $D(s)$ as shown in Fig. 2.8. The feedback can be used to improve the stability, speed up the transient response, improve the steady-state error characteristics, provide disturbance rejection, and decrease the sensitivity to parameter variations.

2.2.1 Stability

The dynamic characteristics of the open-loop system are determined by the poles of $G(s)$ and $D(s)$, that is, the roots of the denominators of $G(s)$ and $D(s)$. Using Eq. (2.10), we can see that the transfer function of the closed-loop system in Fig. 2.8 is

$$\frac{Y(s)}{R(s)} = \frac{D(s)G(s)}{1 + D(s)G(s)} = T(s), \quad (2.19)$$

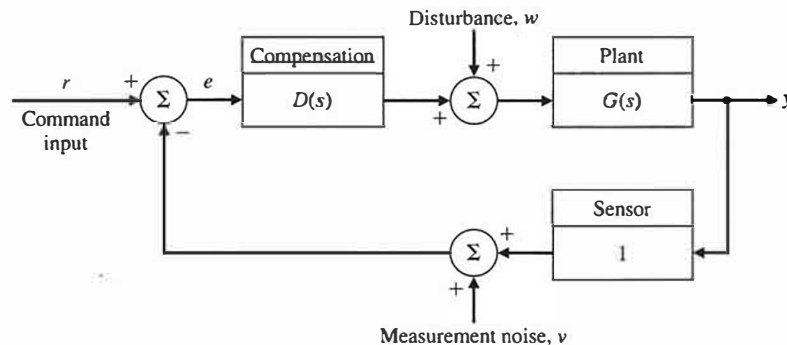
sometimes referred to as the **complementary sensitivity**. In this case, the dynamic characteristics and stability are determined by the poles of the closed-loop transfer function, that is, the roots of

$$1 + D(s)G(s) = 0. \quad (2.20)$$

characteristic equation

This equation is called the **characteristic equation** and is very important in feedback control analysis and design. The roots of the characteristic equation represent the types of motion that will be exhibited by the feedback system. It is clear from Eq. (2.20) that they can be altered at will by the designer via the selection of $D(s)$.

Figure 2.8
A unity feedback system



2.2.2 Steady-State Errors

The difference between the command input r (see Fig. 2.8) and the output y is called the system error, e . Using Eq. (2.10) for the case where the desired output is e , we find that

$$\frac{E(s)}{R(s)} = \frac{1}{1 + D(s)G(s)} = S(s), \quad (2.21)$$

sometimes referred to as the **sensitivity**. For the case where $r(t)$ is a step input and the system is stable, the Final Value Theorem tells us that

$$e_{ss} = \frac{1}{1 + K_p}$$

where

$$K_p = \lim_{s \rightarrow 0} D(s)G(s)$$

and is called the **position-error constant**. If $D(s)G(s)$ has a denominator that does not have s as a factor, K_p and e_{ss} are finite. This kind of system is referred to as **type 0**.

system type

These results can also be seen qualitatively by examining Fig. 2.8. In order for y to be at some desired value ($= r$), the higher the forward loop gain of DG (defined to be K_p), the lower the value of the error, e . An integrator has the property that a zero steady input can produce a finite output, thus producing an infinite gain. Therefore, if there is an integrator in D or G , the steady-state gain will be ∞ and the error will be zero.

Continuing, we define the **velocity constant** as

$$K_v = \lim_{s \rightarrow 0} s D(s)G(s)$$

and the **acceleration constant** as

$$K_a = \lim_{s \rightarrow 0} s^2 D(s)G(s).$$

When K_v is finite, we call the system **type 1**; likewise, when K_a is finite, we call the system **type 2**. For the unity feedback case, it is convenient to categorize the error characteristics for command inputs consisting of steps, ramps, and parabolas. Table 2.1 summarizes the results.

Table 2.1 Errors versus system type for unity feedback

	Step	Ramp	Parabola
Type 0	$\frac{1}{(1+K_p)}$	∞	∞
Type 1	0	$\frac{1}{K_v}$	∞
Type 2	0	0	$\frac{1}{K_a}$

System type can also be defined with respect to the disturbance inputs w . The same ideas hold, but in this case the type is determined by the number of integrators in $D(s)$ only. Thus, if a system had a disturbance as shown in Fig. 2.8 which was constant, the steady-state error e_{ss} of the system would only be zero if $D(s)$ contained an integrator.

2.2.3 PID Control

Proportional, integral, and derivative (PID) control contains three terms. They are proportional control

$$u(t) = Ke(t) \Rightarrow D(s) = K, \quad (2.22)$$

integral control

$$u(t) = \frac{K}{T_i} \int_0^t e(\eta) d\eta \Rightarrow D(s) = \frac{K}{T_i s}, \quad (2.23)$$

and derivative control

$$u(t) = KT_d \dot{e}(t) \Rightarrow D(s) = KT_d s. \quad (2.24)$$

T_i is called the integral (or reset) time, T_d the derivative time, and K the position feedback gain. Thus, the combined transfer function is

$$D(s) = \frac{u(s)}{e(s)} = K \left(1 + \frac{1}{T_i s} + T_d s \right). \quad (2.25)$$

Proportional feedback control can lead to reduced errors to disturbances but still has a small steady-state error. It can also increase the speed of response but typically at the cost of a larger transient overshoot. If the controller also includes a term proportional to the integral of the error, the error to a step can be eliminated as we saw in the previous section. However, there tends to be a further deterioration of the dynamic response. Finally, addition of a term proportional to the error derivative can add damping to the dynamic response. These three terms combined form the classical PID controller. It is widely used in the process industries and commercial controller hardware can be purchased where the user only need “tune” the gains on the three terms.

2.3 Root Locus

The **root locus** is a technique which shows how changes in the system’s open-loop characteristics influence the closed-loop dynamic characteristics. This technique allows us to plot the locus of the closed-loop roots in the s -plane as an open-loop parameter varies, thus producing a root locus. The root locus method is most commonly used to study the effect of the loop gain (K in Eq. (2.25)); however, the method is general and can be used to study the effect of any parameter in $D(s)$

or $G(s)$. In fact, the method can be used to study the roots of any polynomial versus parameters in that polynomial.

A key attribute of the technique is that it allows you to study the **closed-loop roots** while only knowing the factors (poles and zeros) of the **open-loop system**.

2.3.1 Problem Definition

The first step in creating a root locus is to put the polynomials in the **root locus form**

$$1 + K \frac{b(s)}{a(s)} = 0. \quad (2.26)$$

Typically, $Kb(s)/a(s)$ is the open loop transfer function $D(s)G(s)$ of a feedback system; however, this need not be the case. The root locus is the set of values of s for which Eq. (2.26) holds for some real value of K . For the typical case, Eq. (2.26) represents the characteristic equation of the closed-loop system.

The purpose of the root locus is to show in a graphical form the general trend of the roots of a closed-loop system as we vary some parameter. Being able to do this by hand (1) gives the designer the ability to design simple systems without a computer, (2) helps the designer verify and understand computer-generated root loci, and (3) gives insight to the design process.

Equation (2.26) shows that, if K is real and positive, $b(s)/a(s)$ must be real and negative. In other words, if we arrange $b(s)/a(s)$ in polar form as magnitude and phase, then the phase of $b(s)/a(s)$ must be 180° . We thus define the root locus in terms of the phase condition as follows.

root locus definition

180° locus definition: The root locus of $b(s)/a(s)$ is the set of points in the s -plane where the phase of $b(s)/a(s)$ is 180° .

Since the phase is unchanged if an integral multiple of 360° is added, we can express the definition as²

$$\angle \frac{b(s)}{a(s)} = 180^\circ + l360^\circ,$$

where l is any integer. The significance of the definition is that, while it is very difficult to solve a high-order polynomial, computation of phase is relatively easy. When K is positive, we call this the **positive or 180° locus**. When K is real and negative, $b(s)/a(s)$ must be real and positive for s to be on the locus. Therefore, the phase of $b(s)/a(s)$ must be 0° . This case is called the **0° or negative locus**.

² $\angle$ refers to the phase of ().

2.3.2 Root Locus Drawing Rules

The steps in drawing a 180° root locus follow from the basic phase definition. They are

STEP 1 On the s -plane, mark poles (roots of $a(s)$) by an $\times$ and zeros (roots of $b(s)$) by a $\circ$. There will be a branch of the locus departing from every pole and a branch arriving at every zero.

STEP 2 Draw the locus on the real axis to the left of an odd number of real poles plus zeros.

STEP 3 Draw the asymptotes, centered at α and leaving at angles ϕ_l , where

$$n - m = \text{number of asymptotes}$$

$$n = \text{order of } a(s)$$

$$m = \text{order of } b(s)$$

$$\alpha = \frac{\sum p_i - \sum z_i}{n - m} = \frac{-a_1 + b_1}{n - m};$$

$$\phi_l = \frac{180^\circ + (l - 1)360^\circ}{n - m}, \quad l = 1, 2, \dots, n - m.$$

For $n - m > 0$, there will be a branch of the locus approaching each asymptote and departing to infinity. For $n - m < 0$, there will be a branch of the locus arriving from infinity along each asymptote.

STEP 4 Compute locus departure angles from the poles and arrival angles at the zeros where

$$q\phi_{dep} = \sum \psi_i - \sum \phi_i - 180^\circ - l360^\circ$$

$$q\psi_{arr} = \sum \phi_i - \sum \psi_i + 180^\circ + l360^\circ$$

where q is the order of the pole or zero and l takes on q integer values so that the angles are between $\pm 180^\circ$. ψ_i is the angle of the line going from the i_{th} pole to the pole or zero whose angle of departure or arrival is being computed. Similarly, ϕ_i is the angle of the line from the i_{th} zero.

STEP 5 If further refinement is required at the stability boundary, assume $s_0 = j\omega_0$ and compute the point(s) where the locus crosses the imaginary axis for positive K .

STEP 6 For the case of multiple roots, two loci come together at 180° and break away at $\pm 90^\circ$. Three loci segments approach each other at angles of 120° and depart at angles rotated by 60° .

STEP 7 Complete the locus, using the facts developed in the previous steps and making reference to the illustrative loci for guidance. The loci branches start at poles and end at zeros or infinity.

STEP 8 Select the desired point on the locus that meets the specifications (s_o), then use the magnitude condition from Eq. (2.26) to find that the value of K associated with that point is

$$K = \frac{1}{|b(s_o)/a(s_o)|}.$$

When K is negative, the definition of the root locus in terms of the phase relationship is

0° locus definition: The root locus of $b(s)/a(s)$ is the set of points in the s -plane where the phase of $b(s)/a(s)$ is 0° .

For this case, the Steps above are modified as follows

STEP 2 Draw the locus on the real axis to the left of an *even* number of real poles plus zeros.

STEP 3 The asymptotes depart at

$$\phi_l = \frac{(l-1)360^\circ}{n-m}, \quad l = 1, 2, \dots, n-m.$$

STEP 4 The locus departure and arrival angles are modified to

$$\begin{aligned} q\phi_{dep} &= \sum \psi_i - \sum \phi_i - l360^\circ \\ q\psi_{arr} &= \sum \phi_i - \sum \psi_i + l360^\circ. \end{aligned}$$

Note that the 180° term has been removed.

◆ Example 2.1 Root Locus Sketch

Sketch the root locus versus K (positive and negative) for the case where the open-loop system is given by

$$G(s) = K \frac{s}{s^2 + 1}.$$

Solution. First let's do the 180° locus.

STEP 1: There is a zero at $s = 0$ and poles at $s = \pm j\omega$.

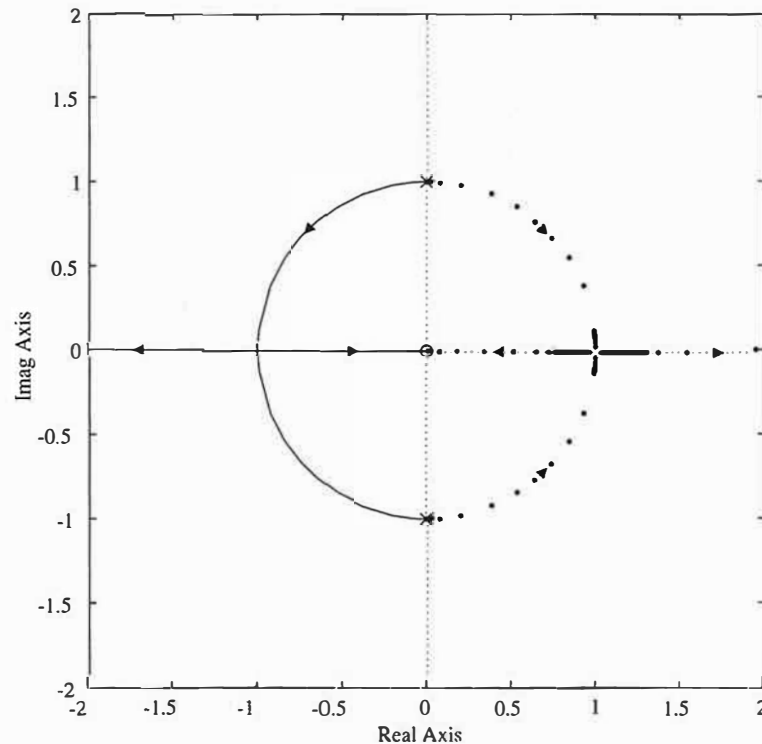
STEP 2: There is a locus on the entire negative real axis.

STEP 3: $n - m = 1$, therefore, there is one asymptote and it departs at 180° ; that is, along the negative real axis.

STEP 4: The departure angle at the upper pole is calculated as

$$\phi_1 = 90^\circ - 90^\circ - 180^\circ = -180^\circ,$$

Figure 2.9
Example root locus
sketch



thus, the locus departs from the upper pole horizontally to the left. The departure angle from the lower pole also turns out to be -180° and that branch of the locus also departs horizontally to the left.

We know that there is a locus segment along the entire negative real axis; however, we also know that there is a locus branch moving to the right and arriving at the zero, and that there is a branch departing to infinity at the far left. Therefore, the two branches from the poles must join the real axis at some point and split in opposite directions. It turns out that the two complex branches form a semi-circle as they approach the real axis. The solid lines in Fig. 2.9 show the sketch of this 180° locus.

For the 0° locus, there is a segment along the positive real axis and the angles of departure are both 0° . The result is shown in the figure by the dotted lines.

2.3.3 Computer-Aided Loci

The most common approach to machine computation of the root locus is to cast the problem as a polynomial in the form $a(s) + Kb(s) = 0$, and, for a sequence of values of K varying from near zero to a large value, solve the polynomial for

its n roots by any of many available numerical methods. A disadvantage of this method is that the resulting root locations are very unevenly distributed in the s -plane. For example, near a point of multiple roots, the sensitivity of the root locations to the parameter value is very great, and the roots just fly through such points, the plots appear to be irregular, and sometimes important features are missed. As a result, it is useful to have the root locus plotting rules in mind when interpreting computer plots. The polynomial is generally solved by transforming the problem to state-variable form and using the QR algorithm which solves for the eigenvalues of the closed-loop system matrix.

◆ Example 2.2 CACSD Root Locus

1. Plot the root locus using MATLAB for the open-loop system shown in Fig. 2.8 with

$$G(s) = \frac{10}{s(s+2)}, \quad \text{and} \quad D(s) = K \frac{s+3}{s+10}.$$

2. Find the gain K associated with the point of maximum damping of the complex roots for the given $D(s)$ and plot the step response with that value of K .
3. Reconcile the root locus plot with the hand plotting rules and compare the computer-based step response with the rules of thumb in Eqs. (2.16)–(2.18).

Solution.

1. The MATLAB script following will generate the desired locus plot which is shown in Fig. 2.10(a).

```
s=tf('s');
sysG=10/(s^2+2*s);
sysD=(s+3)/(s+10);
sys=sysG*sysD;
rlocus(sys)
```

2. The statement

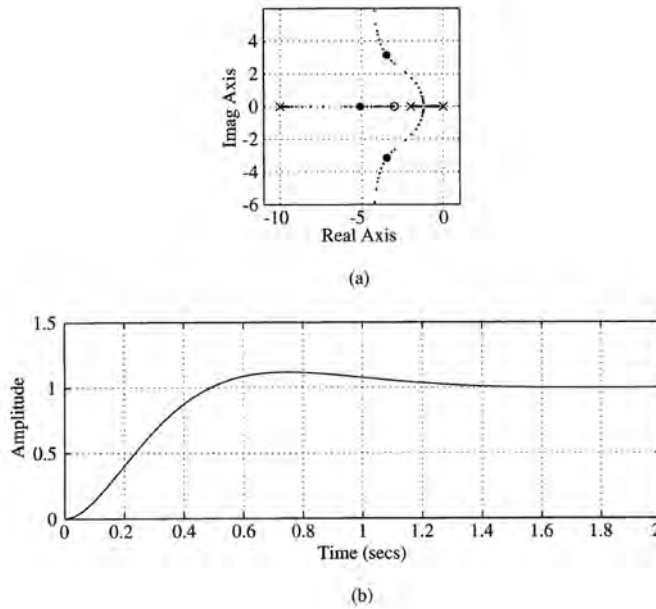
```
[K,p] = rlocfind(sys)
```

will place a cross-hair on the plot which can be moved with the mouse to the desired point on the locus in order to find the associated gain K and pole locations p . Given this value of K , ($\cong 3.7$) the script

```
K = 3.7
sysCL = feedback(K*sys,1)
step(sysCL)
```

produces the desired step response shown in Fig. 2.10(b).

Figure 2.10
Example of CACSD for
(a) root locus and (b)
step response



3. The root locus in Fig. 2.10(a) has locus segments to the left of odd numbers of poles and zeros (Step 2); has two asymptotes departing at $\pm 90^\circ$ and centered at

$$\alpha = \frac{-2 + 3 - 10}{2} = -4.5,$$

(Step 3); and has branches departing the real axis at the multiple roots between the two poles at $\pm 90^\circ$ (Step 6). The gain associated with the desired root at $s = -3.5 \pm j3.1$ can be determined from Step 8 by

$$K = \frac{(4.7)(3.5)(7.2)}{(3.2)(10)} = 3.7$$

where 4.7 is the distance from the root to the pole at $s = 0$; 3.5 is the distance to the pole at $s = -2$; 7.2 is the distance to the pole at $s = -10$; 3.2 is the distance to the zero at $s = -3$; and 10 is from the gain of $G(s)$.

The step response shown in Fig. 2.10(b) has $t_r \cong 0.4$ sec, $t_s \cong 1.4$ sec, and $M_p \cong 10\%$. The closed-loop roots with $K = 3.7$ are at $s = -5.1, -3.5 \pm j3.1$; thus, for the complex roots, $\zeta = 0.74$, $\omega_n = 4.7$ rad/sec, and $\sigma = 3.5$. The predicted values given in Section 2.1.7 suggest that

$$t_r \approx \frac{1.8}{\omega_n} = 0.38 \text{ sec}$$

$$t_s \approx \frac{4.6}{\sigma} = 1.3 \text{ sec}$$

$$M_p \approx e^{-\pi\zeta/\sqrt{1-\zeta^2}} = 4\% \text{ (Fig. 2.7)}$$

The predicted values based on the second order system with no zeros predict a t_r and t_s that are a little faster than the plot due to the presence of the extra pole. The predicted M_p of 4% is much smaller than the 10% shown on Fig. 2.10(b), also a result of the extra zero. If a lower overshoot was required, the zero would need to be moved to the right as needed.

2.4 Frequency Response Design

The response of a linear system to a sinusoidal input is referred to as the system's **frequency response**. A system described by

$$\frac{Y(s)}{U(s)} = G(s),$$

where the input $u(t)$ is a sine wave with an amplitude of U_o and frequency ω

$$u(t) = U_o \sin \omega_1 t,$$

which has a Laplace transform

$$U(s) = \frac{U_o \omega_1}{s^2 + \omega_1^2},$$

has a response with the transform,

$$Y(s) = G(s) \frac{U_o \omega_1}{s^2 + \omega_1^2}. \quad (2.27)$$

A partial fraction expansion of Eq. (2.27) will result in terms that represent the natural behavior of $G(s)$ and terms representing the sinusoidal input. Providing that all the natural behavior is stable, those terms will die out and the only terms left in the steady state are those due to the sinusoidal excitation, that is

$$Y(s) = \dots + \frac{\alpha_o}{s + j\omega_1} + \frac{\alpha_o^*}{s - j\omega_1} \quad (2.28)$$

where α_o and α_o^* would be found by performing the partial fraction expansion. After the natural transients have died out, the time response is

$$y(t) = 2|\alpha_o| \sin(\omega_1 t + \phi) = U_o A \sin(\omega_1 t + \phi)$$

where

$$A = |G(j\omega_1)| = |G(s)| \Big|_{s=j\omega_1}, \quad (2.29)$$

$$\phi = \tan^{-1} \frac{\text{Im}[G(j\omega_1)]}{\text{Re}[G(j\omega_1)]} = \angle G(j\omega_1). \quad (2.30)$$

So, a stable linear system $G(s)$ excited by a sinusoid will eventually exhibit a sinusoidal output y with the same frequency as the input u . The **magnitude**, $A(\omega_1)$ of y with respect to the input, $= |G(j\omega_1)|$ and the **phase**, $\phi(\omega_1)$, is $\angle G(j\omega_1)$; that is, the magnitude and phase of $G(s)$ is evaluated by letting s take on values along the imaginary ($j\omega$) axis. In addition to the response to a sinusoid, the analysis of the frequency response of a system is very useful in the determination of stability of a closed-loop system given its open-loop transfer function.

A key reason that the frequency response is so valuable is that the designer can determine the frequency response experimentally with no prior knowledge of the system's model or transfer function. The system is excited by a sinusoid with varying frequency and the magnitude $A(\omega)$ is obtained by a measurement of the ratio of the output sinusoid to input sinusoid in the steady-state at each frequency. The phase $\phi(\omega)$ is the measured difference in phase between input and output signals. As an example, frequency responses of the second-order system

$$G(s) = \frac{1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1}$$

are plotted for various values of ζ in Fig. 2.11 which is done by MATLAB with `bode(sys)`.

2.4.1 Specifications

bandwidth

A natural specification for system performance in terms of frequency response is the **bandwidth**, defined to be the maximum frequency at which the output of a system will track an input sinusoid in a satisfactory manner. By convention, for the system shown in Fig. 2.12 with a sinusoidal input r , the bandwidth is the frequency of r at which the output y is attenuated to a factor of 0.707 times the input (or down 3 dB). Figure 2.13 depicts the idea graphically for the frequency response of the *closed-loop* transfer function (defined to be $T(s)$ in Eq. (2.19))

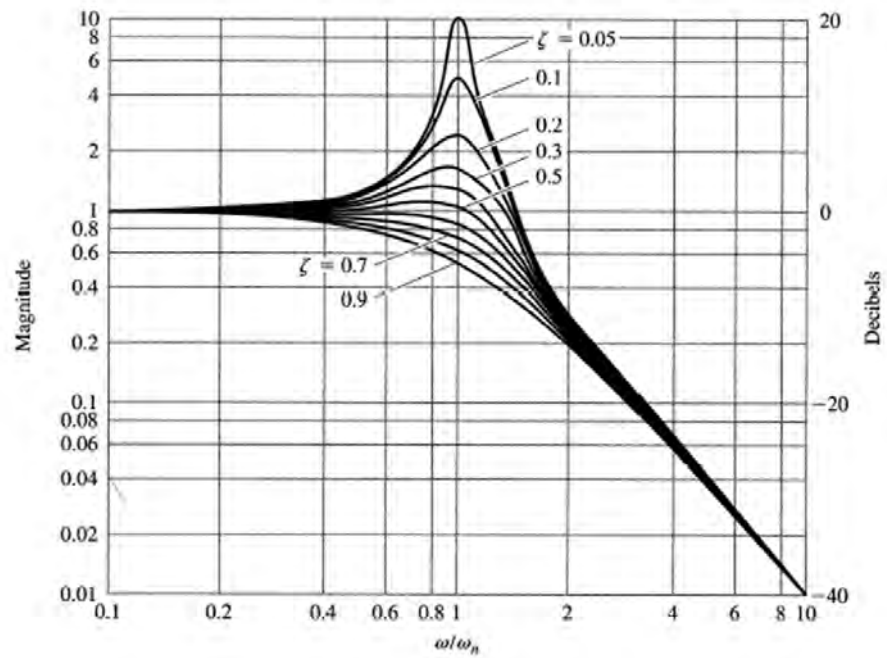
$$\frac{Y(s)}{R(s)} = T(s) = \frac{KG(s)}{1 + KG(s)}.$$

The plot is typical of most closed-loop systems in that 1) the output follows the input, $|T| \cong 1$, at the lower excitation frequencies, and 2) the output ceases to follow the input, $|T| < 1$, at the higher excitation frequencies.

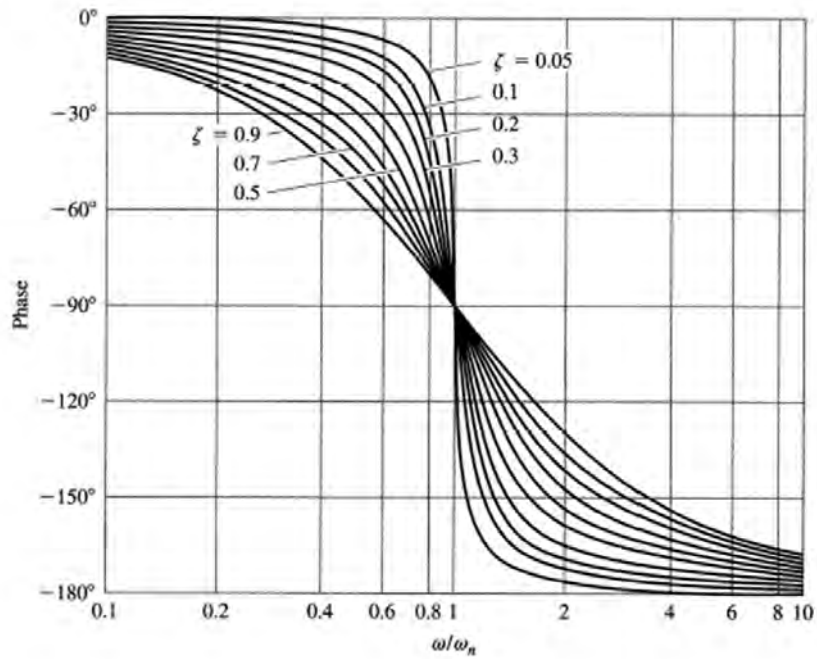
The bandwidth ω_{BW} is a measure of the speed of response and is therefore similar to the time-domain measure of rise time t_r or the s -plane measure of natural frequency ω_n . In fact, it can be seen from Fig. 2.11 that the bandwidth will be equal to the natural frequency when $\zeta = 0.7$. For other damping ratios, the bandwidth is approximately equal to the natural frequency with an error typically less than a factor of 2.

The resonant peak M_r is a measure of the damping, as evidenced by Fig. 2.11 where the peak is approximately the value at $\omega = \omega_n$, which is $\frac{1}{2\zeta}$ for $\zeta < 0.5$.

Figure 2.11
 (a) Magnitude and (b)
 phase of a second-order
 system



(a)



(b)

Figure 2.12
Simplified system
definition

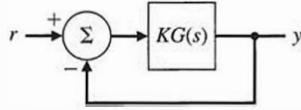
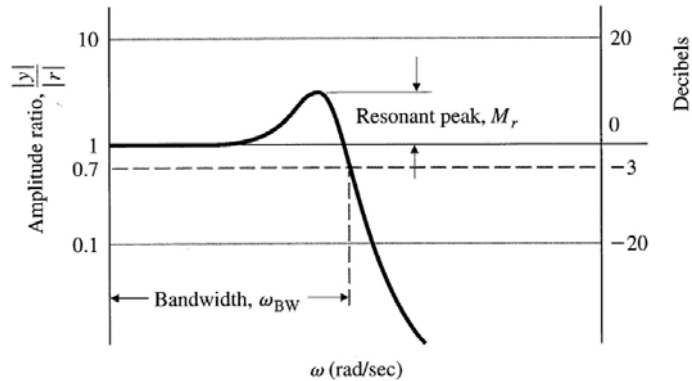


Figure 2.13
Definitions of bandwidth
and resonant peak



2.4.2 Bode Plot Techniques

It is useful to be able to plot the frequency response of a system by hand in order to (a) design simple systems without the aid of a computer, (b) check computer-based results, and (c) understand the effect of compensation changes in design iterations. H. W. Bode developed plotting techniques in the 1930s that enabled quick hand plotting of the frequency response. His rules are:

STEP 1 Manipulate the transfer function into the **Bode form**

$$KG(j\omega) = K_o(j\omega)^n \frac{(j\omega\tau_1 + 1)(j\omega\tau_2 + 1) \cdots}{(j\omega\tau_a + 1)(j\omega\tau_b + 1) \cdots}$$

STEP 2 Determine the value of n for the $K_o(j\omega)^n$ term. Plot the low-frequency magnitude asymptote through the point K_o at $\omega = 1$ rad/sec with a slope of n (or $n \times 20$ dB per decade).

STEP 3 Determine the **break points** where $\omega = 1/\tau_i$. Complete the composite magnitude asymptotes by extending the low frequency asymptote until the first frequency break point, then stepping the slope by ± 1 or ± 2 , depending on whether the break point is from a first or second order term in the numerator or denominator, and continuing through all break points in ascending order.

STEP 4 Sketch in the approximate magnitude curve by increasing from the asymptote by a factor of 1.4 (+3 dB) at first order numerator breaks and decreasing it by a factor of 0.707 (−3 dB) at first order denominator breaks. At second order break points, sketch in the resonant peak (or valley) according to Fig. 2.11(a) using the relation that $|G(j\omega)| = 1/(2\zeta)$ at the break.

STEP 5 Plot the low frequency asymptote of the phase curve, $\phi = n \times 90^\circ$.

STEP 6 As a guide, sketch in the approximate phase curve by changing the phase gradually over two decades by $\pm 90^\circ$ or $\pm 180^\circ$ at each break point in ascending order. For first order terms in the numerator, the gradual change of phase is $+90^\circ$; in the denominator, the change is -90° . For second order terms, the change is $\pm 180^\circ$.

STEP 7 Locate the asymptotes for each individual phase curve so that their phase change corresponds to the steps in the phase from the approximate curve indicated by Step 6. Sketch in each individual phase curve as indicated by Fig. 2.14 or Fig. 2.11(b).

STEP 8 Graphically add each phase curve. Use dividers if an accuracy of about $\pm 5^\circ$ is desired. If lessor accuracy is acceptable, the composite curve can be done by eye, keeping in mind that the curve will start at the lowest frequency asymptote and end on the highest frequency asymptote, and will approach the intermediate asymptotes to an extent that is determined by the proximity of the break points to each other.

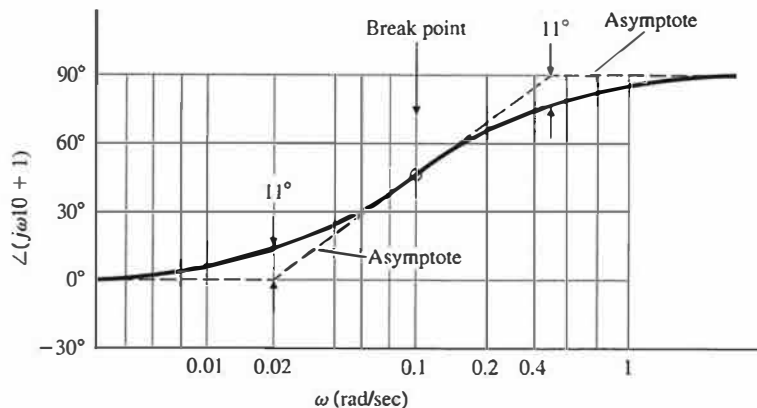
2.4.3 Steady-State Errors

Section 2.2.2 showed that the steady-state error of a feedback system decreases as the gain of the open loop transfer function increases. In plotting a composite magnitude curve, the low frequency asymptote is given by

$$KG(j\omega) = K_o(j\omega)^n. \quad (2.31)$$

Therefore, we see that the larger the value of the magnitude on the low-frequency asymptote, the lower the steady-state errors will be for the closed loop system. This idea is very useful in designing compensation.

Figure 2.14
Phase plot for $j\omega\tau + 1$;
 $\tau = 0.1$



For a system with $n = 0$, (a type 0 system) the low frequency asymptote is a constant and the gain K_o of the open loop system is equal to the position error constant, K_p . For a system where $n = -1$, (a type 1 system) the low frequency asymptote has a slope of -1 and $K_v = K_o$.

The easiest way of determining the value of K_v in a type 1 system is to read the magnitude of the low frequency asymptote at a frequency low enough to be well below the any of the break points because $\frac{K_v}{\omega}$ equals the magnitude at these frequencies. In some cases, the lowest frequency break point will be below $\omega = 1$ rad/sec, therefore the asymptote can be extended to $\omega = 1$ rad/sec in order to read K_v directly.

2.4.4 Stability Margins

If the closed-loop transfer function of a system is known, the stability of the system can be determined by simply inspecting the denominator in factored form to observe whether the real parts are positive or negative. However, the closed-loop transfer function is not usually known; therefore, we would like to determine closed-loop stability by evaluating the frequency response of the *open-loop* transfer function $KG(j\omega)$ and then performing a simple test on that response. This can be done without a math model of the system by experimentally determining the open-loop frequency response.

We saw in Section 2.3.1 that all points on the root locus have the property that

$$|KG(s)| = 1 \quad \text{and} \quad \angle(KG(s)) = 180^\circ.$$

At the point of neutral stability we see that these root-locus conditions hold for $s = j\omega$, so

$$|KG(j\omega)| = 1 \quad \text{and} \quad \angle(KG(j\omega)) = 180^\circ. \quad (2.32)$$

Thus a Bode plot of a system that is neutrally stable (that is, with the value of K such that the closed-loop roots fall on the imaginary axis) will satisfy the conditions of Eq. (2.32). That means that the magnitude plot must equal 1 at the same frequency that the phase plot equals 180° . Typically, a system becomes less stable as the gain increases; therefore, we have the condition for stability

$$|KG(j\omega)| < 1 \quad \text{at} \quad \angle(KG(j\omega)) = -180^\circ. \quad (2.33)$$

This stability criterion holds for all systems where increasing gain leads to instability and $|KG(j\omega)|$ crosses the magnitude = 1 once, the most common situation. However, there are systems where an increasing gain can lead from instability to stability and in this case, the stability condition is

$$|KG(j\omega)| > 1 \quad \text{at} \quad \angle(KG(j\omega)) = -180^\circ.$$

One way that will frequently resolve the ambiguity is to perform a rough sketch of the root locus to resolve the question of whether increasing gain leads to stability or instability. The rigorous way to resolve the ambiguity is to use the Nyquist stability criterion, which is reviewed in Section 7.5.1 for continuous systems.

gain margin
phase margin

Two quantities that measure the stability margin of a system are directly related to the stability criterion of Eq. (2.33): gain margin and phase margin. The **gain margin** (GM) is the factor by which the gain is less than the neutral stability value when the phase = 180° . The **phase margin** (PM) is the amount by which the phase of $G(s)$ exceeds -180° when $|KG(j\omega)| = 1$. The two margins are alternate ways of measuring the degree to which the stability conditions of Eq. (2.33) are met.

The phase margin is generally related to the damping of a system. For a second-order system, the approximation that

$$\zeta \cong \frac{\text{PM}}{100}$$

is commonly used. Therefore, if it were known that a system was to be designed using frequency response methods, it would make sense to specify the speed of response of the system in terms of a required bandwidth and the stability of the system in terms of a required phase margin.

2.4.5 Bode's Gain-Phase Relationship

One of Bode's important contributions is his theorem that states

For any minimum phase system (that is, one with no time delays, RHP zeros *or* poles), the phase of $G(j\omega)$ is uniquely related to the integral of the magnitude of $G(j\omega)$.

When the slope of $|G(j\omega)|$ versus ω on a log-log scale persists at a constant value for nearly a decade of frequency, the relationship is particularly simple

$$\angle G(j\omega) \cong n \times 90^\circ, \quad (2.34)$$

where n is the slope of $|G(j\omega)|$ in units of decade of amplitude per decade of frequency.

Equation (2.34) is used as a guide to infer stability from $|G(j\omega)|$ alone. When $|KG(j\omega)| = 1$, the **crossover frequency**, the phase

$$\begin{aligned} \angle G(j\omega) &\cong -90^\circ && \text{if } n = -1, \\ \angle G(j\omega) &\cong -180^\circ && \text{if } n = -2. \end{aligned}$$

For stability we want $\angle G(j\omega) > -180^\circ$ for a PM > 0 . Therefore we adjust the $|KG(j\omega)|$ curve so that it has a slope of -1 at the crossover frequency. If the slope is -1 for a decade above and below the crossover frequency, the PM would be approximately 90° ; however, to ensure a reasonable PM, it is usually only

necessary to insist on a -1 slope (-20 dB per decade) persisting for a decade in frequency that is centered at the crossover frequency.

2.4.6 Design

One of the very useful aspects of frequency-response design is the ease with which we can evaluate the effects of gain changes. In fact, we can determine the PM for any value of K without redrawing the magnitude or phase information. We need only indicate on the figure where $|KG(j\omega)| = 1$ for selected trial values of K since varying K has the effect of sliding the magnitude plot up or down.

◆ Example 2.3 Frequency-Response Design

For a plant given by

$$G(s) = K \frac{1}{s(s+1)^2},$$

determine the PM and GM for the system with unity feedback and (a) $K = 1$, (b) determine if the system is stable if $K = 5$, and (c) find what value of K is required to achieve a PM of (i) 45° , and (ii) 70° .

Solution.

Using the hand plotting rules, we see that the low frequency asymptote has a slope of -1 and goes thru magnitude $= 1$ at $\omega = 1$ rad/sec. The slope changes to -3 at the break point ($\omega = 1$). We can then sketch in the actual magnitude curve, noting (STEP 4 in Section 2.4.2) that it will go below the asymptote intersection by -6 dB because there is a slope change of -2 at that break point. The curve is sketched in Fig. 2.15. The phase curve starts out at -90° and drops to -270° along the asymptote as sketched in the figure according to STEP 7.

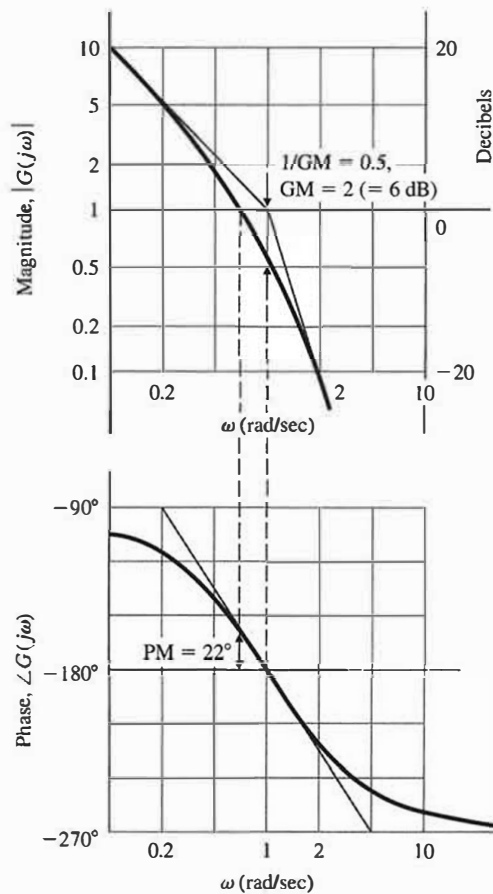
Using MATLAB, the statements

```
num = 1, den = [1 2 1 0]
sys = tf(num,den)
bode(sys).
```

will also create the plots of magnitude and phase for this example. The curves are drawn in Fig. 2.15 showing the PM and GM for $K = 1$ and the same curves are drawn in Fig. 2.16 showing the PM's for $K = 5, 0.5$, & 0.2 .

- (a) We can read the PM from Fig. 2.15 to be 22° .
- (b) Fig. 2.16 shows that the system is unstable for $K=5$.
- (c) (i) PM $= 45^\circ$ when $K = 0.5$, and (ii) PM $= 70^\circ$ when $K = 0.2$

Figure 2.15
Magnitude and phase
plots with PM and GM
for $1/s(s+1)^2$



2.5 Compensation

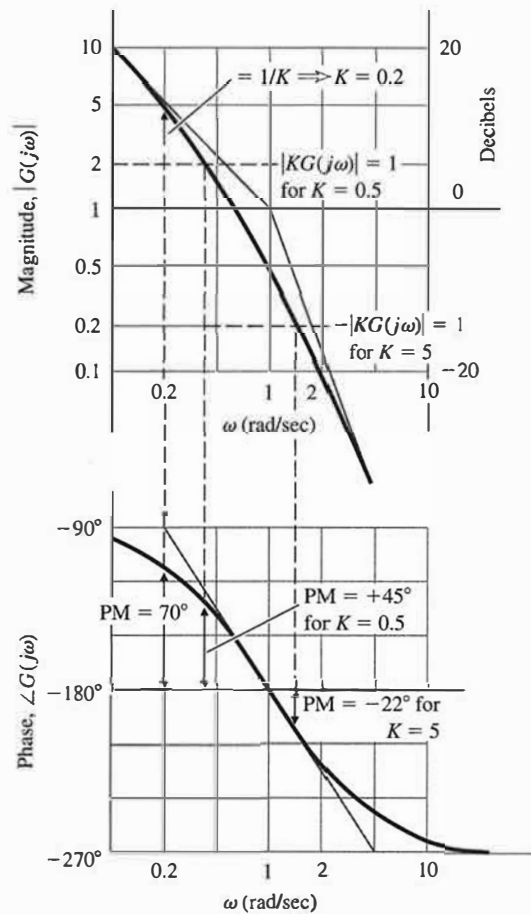
If the plant dynamics are of such a nature that a satisfactory design cannot be achieved by adjustment of the feedback gain alone, then some modification or **compensation** must be made in the feedback to achieve the desired specifications. Typically, it takes the form

$$D(s) = K \frac{s + z}{s + p}$$

lead compensation
lag compensation

where it is called **lead compensation** if $z < p$ and **lag compensation** if $z > p$. Lead compensation approximates the addition of a derivative control term and tends to increase the bandwidth and the speed of response while decreasing the overshoot. Lag compensation approximates integral control and tends to improve the steady-state error.

Figure 2.16
PM versus K for
 $1/(s+1)^2$



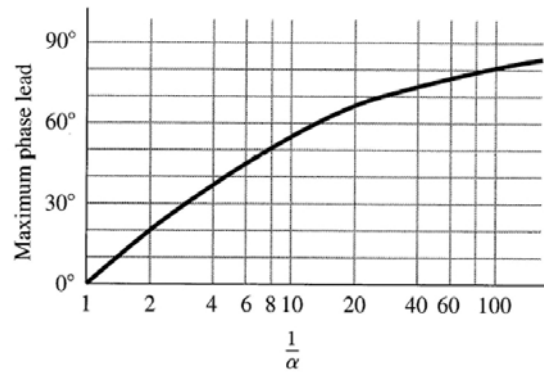
The design of lead compensation typically entails placing the zero z at a frequency that is lower than the magnitude = 1 crossover frequency and the pole higher than the crossover frequency. Lead compensation provides an increased magnitude slope and an increased phase in the interval between these two break points; the maximum being halfway between the two break points on a logarithmic scale. The maximum phase increase is

$$\delta\phi = \sin^{-1} \frac{1 - \alpha}{1 + \alpha} \quad \text{where} \quad \alpha = \frac{z}{p}$$

and is plotted versus α in Fig. 2.17.

The design of lag compensation typically entails placing both break points well below the crossover frequency. Lag compensation decreases the phase in the vicinity of the two break points; therefore, z should be well below the crossover

Figure 2.17
Maximum phase
increase for lead
compensation



frequency in order to prevent the compensation from degrading the PM and the system stability. The primary role of lag compensation is to increase the gain (magnitude of the frequency response) at the low frequencies. As we saw in Section 2.4.3, this will decrease the steady-state error.

2.6 State-Space Design

We saw in Section 2.1.1 that equations of motion could be written in the state-variable form of Eqs. (2.1) and (2.2). The **state-space** design approach utilizes this way of describing the plant and arrives directly with feedback controllers (compensation) without the need to determine transforms. Advantages of state-space design are especially apparent when the system to be controlled has more than one control input or more than one sensed output, called multivariable or multi input-multi output (MIMO). However, we will review only the single input-single output (SISO) case here. For readers not familiar with state-space design, the material in this section is not required for comprehension of the remainder of the book. The basic ideas of state-space design are covered in detail in Chapter 8 for the discrete case and that chapter does not require that the reader be knowledgeable about continuous state-space design. Chapter 9 extends state-space design for discrete systems to optimal control design for the multivariable case.

One of the attractive features of the state-space design method is that it consists of a sequence of independent steps. The first step, discussed in Section 2.6.1, is to determine the control. The purpose of the control law is to allow us to design a set of pole locations for the closed-loop system that will correspond to satisfactory dynamic response in terms of rise-time, overshoot, or other measures of transient response.

The second step—necessary if the full state is not available—is to design an **estimator** (sometimes called an **observer**), which computes an estimate of the

estimator
observer

entire state vector when provided with the measurements of the system indicated by Eq. (2.2). We review estimator design in Section 2.6.2.

The third step consists of combining the control law and the estimator. Figure 2.18 shows how the control law and the estimator fit together and how the combination takes the place of what we have been previously referring to as compensation.

The fourth and final step is to introduce the reference input in such a way that the plant output will track external commands with acceptable rise-time, overshoot and settling time values. Figure 2.18 shows the command input r introduced in the same relative position as was done with the transform design methods; however, in Section 2.6.4 we will show how to introduce the reference input in a different way that results in a better system response.

2.6.1 Control Law

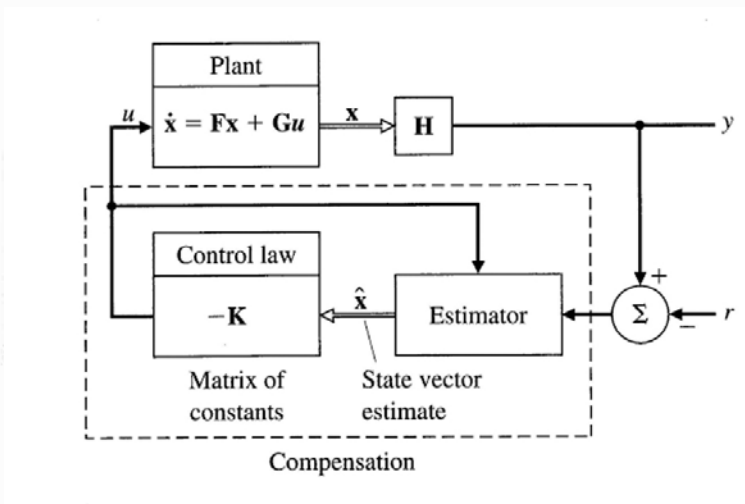
The first step is to find the control law as feedback of a linear combination of all the state variables—that is,

$$u = -\mathbf{K}\mathbf{x} = -[K_1 \ K_2 \ \cdots \ K_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}. \quad (2.35)$$

We *assume* for design purposes that all the elements of the state vector are at our disposal, an infrequent situation for an actual system, but an expedient assumption for the time being.

For an n th-order system there will be n feedback gains $K_1, \dots, K_n$, and since there are n roots (or *poles*) of the system, it is possible that there are enough degrees of freedom to select *arbitrarily* any desired root location by choosing the proper values of K_i .

Figure 2.18
Schematic diagram of
state-space design
elements



Substituting the feedback law, Eq. (2.35) into the system described by Eq. (2.1) yields

$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} - \mathbf{G}\mathbf{K}\mathbf{x}. \quad (2.36)$$

The characteristic equation of this closed-loop system is

$$\det[s\mathbf{I} - (\mathbf{F} - \mathbf{G}\mathbf{K})] = 0. \quad (2.37)$$

When evaluated, this yields an n th-order polynomial in s containing the gains $K_1, \dots, K_n$. The control-law design then consists of picking the gains $\mathbf{K}$ so that the roots of Eq. (2.37) are in desirable locations. Selection of desirable locations for the roots is an inexact science and may require some iteration by the designer. For now, we will assume that the desired locations are known, say

$$s = s_1, s_2, \dots, s_n.$$

Then the corresponding desired (control) characteristic equation is

$$\alpha_c(s) = (s - s_1)(s - s_2) \cdots (s - s_n) = 0. \quad (2.38)$$

Hence the required elements of $\mathbf{K}$ are obtained by matching coefficients in Eq. (2.37) and Eq. (2.38). This forces the system characteristic equation to be identical with the desired characteristic equation and the closed-loop poles to be placed at the desired locations.

controllability

The calculation of $\mathbf{K}$ can be done providing the system is **controllable**. Systems that are not controllable have certain modes or subsystems that are unaffected by the control. This usually means that parts of the system are physically disconnected from the input. Although there is a mathematical test for controllability, it is good practice to insist on the stronger condition that the control input be as strongly coupled to the modes of interest as possible.

It is theoretically possible to solve for $\mathbf{K}$ by hand with Eq. (2.37) and Eq. (2.38). In practice, this is almost never done. Ackermann's formula for this calculation has been implemented in MATLAB as the function `acker.m` and can be used for the design of SISO systems with a small (≤ 10) number of state variables. For more complex cases a more reliable formula is available, implemented in MATLAB as the function `place`. A modest limitation on `place.m` is that none of the desired closed-loop poles are repeated; i.e., that the poles are *distinct*, a requirement that does not apply to `acker`. Both `acker` and `place` require inputs consisting of the system description matrices, $\mathbf{F}$ and $\mathbf{G}$, and a vector, $\mathbf{p}$, of n desired pole locations. Their output is the feedback gain $\mathbf{K}$. Thus the MATLAB statements

$$\mathbf{K} = \text{acker}(\mathbf{F}, \mathbf{G}, \mathbf{p}) \quad \text{or} \quad \mathbf{K} = \text{place}(\mathbf{F}, \mathbf{G}, \mathbf{P})$$

will provide the desired value of $\mathbf{K}$. When selecting the desired root locations, it is always useful to keep in mind that the control effort required is related to how far the open-loop poles are moved by the feedback. Furthermore, when a zero is near a pole, the system may be nearly uncontrollable and moving such a pole may

require large control effort. Therefore, a pole placement philosophy that aims to fix only the undesirable aspects of the open-loop response and avoids either large increases in bandwidth or efforts to move poles that are near zeros will typically allow smaller gains and thus smaller control actuators.

LQR

A method called the **linear quadratic regulator (LQR)** specifically addresses the issue of achieving a balance between good system response and the control effort required. The method consists of calculating the gain K that minimizes a **cost function**

optimal control

$$\mathcal{J} = \int_0^{\infty} [\mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{u}^T \mathbf{R} \mathbf{u}] dt \quad (2.39)$$

where $\mathbf{Q}$ is an $n \times n$ state weighting matrix, $\mathbf{R}$ is an $m \times m$ control weighting matrix, and m is the number of control inputs in a multi-input system. For the SISO systems that we are primarily concerned with here, $m = 1$ and $\mathbf{R}$ is a scalar R . The weights $\mathbf{Q}$ and R are picked by the designer by trial-and-error in order to arrive at the desired balance between state errors $\mathbf{x}^T \mathbf{x}$ and control usage u^2 , thus avoiding the necessity of picking desired pole locations that do not use excessive control. Generally, $\mathbf{Q}$ is a diagonal matrix with a weighting factor on one or more of the state-vector elements while $R = 1$. It is perfectly acceptable to only weight one element, in fact, the element representing the system output is often the only element weighted. Rules of thumb that help in picking the weights are that (1) the bandwidth of the system increases as overall values in $\mathbf{Q}$ increase, (2) the damping increases as the term in $\mathbf{Q}$ that weights the velocity type state elements increase, and (3) a portion of a system can be made faster by increasing the weights on the state elements representing that portion of the system. The MATLAB statement

$$\mathbf{K} = \text{lqr}(\mathbf{F}, \mathbf{G}, \mathbf{Q}, \mathbf{R})$$

solves for the $\mathbf{K}$ that minimizes the cost, $\mathcal{J}$.

◆ Example 2.4 State-Space Control Design

For a plant given by

$$G(s) = \frac{1}{s^2},$$

- (a) Find the feedback gain matrix $\mathbf{K}$ that yields closed-loop roots with $\omega_n = 3$ rad/sec and $\zeta = 0.8$.
- (b) Investigate the roots obtained by using LQR with

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 100 & 0 \\ 0 & 0 \end{bmatrix}, \quad \text{and} \quad \begin{bmatrix} 100 & 0 \\ 0 & 5 \end{bmatrix}$$

and $R = 1$.

Solution. The state-variable description of $G(s)$ is (Eq. (2.4) with $\omega_o = 0$, $\zeta = 0$, and $K_o = 1$)

$$\mathbf{F} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ \mathbf{H} = [1 \quad 0], \quad J = 0.$$

(a) The desired characteristic equation is

$$\alpha_c(s) = s^2 + 2\zeta\omega_n s + \omega_n^2 = 0.$$

Therefore, the MATLAB script

```
F = [0 1; 0 0]
G = [0; 1]
Wn = 3
Ze = 0.8
p = roots([1 2*Wn*Ze Wn^2])
K = acker(F,G,p)
```

provides the answer

$$\mathbf{K} = [9 \quad 4.8].$$

(b) The scripts

```
Q = [1 0; 0 0], [100 0; 0 0], and [100 0; 0 5]
R = 1
K = lqr(F,G,Q,R)
p = eig(F - G*K)
[Wn,Ze] = damp(p)
```

compute feedback gains of

$$\mathbf{K} = [1 \quad 1.4], [10 \quad 4.5], \text{ and } [10 \quad 5].$$

which produces natural frequencies of

$$\omega_n = 1, \quad 3.2, \quad \text{and} \quad 3.2 \text{ rad/sec}$$

and damping of

$$\zeta = 0.71, \quad 0.71, \quad \text{and} \quad 0.79.$$

For this simple example, use of `acker` is the easier way to find $\mathbf{K}$; however, in more complex systems with higher order roots, it is easier to use `lqr` rather than iterate on the best value for all the roots.

2.6.2 Estimator Design

For a system described by Eqs. (2.1) and (2.2), an estimate, $\hat{\mathbf{x}}$, of the full state vector, $\mathbf{x}$, can be obtained based on measurements of the output, y , from

$$\dot{\hat{\mathbf{x}}} = \mathbf{F}\hat{\mathbf{x}} + \mathbf{G}u + \mathbf{L}(y - \mathbf{H}\hat{\mathbf{x}}). \quad (2.40)$$

Here $\mathbf{L}$ is a proportional gain defined as

$$\mathbf{L} = [l_1, l_2, \dots, l_n]^T, \quad (2.41)$$

and is chosen to achieve satisfactory error characteristics. The dynamics of the error can be obtained by subtracting the estimate (Eq. 2.40) from the state (Eq. 2.1), to get the error equation

$$\dot{\tilde{\mathbf{x}}} = (\mathbf{F} - \mathbf{LH})\tilde{\mathbf{x}}. \quad (2.42)$$

The characteristic equation of the error is now given by

$$\det[s\mathbf{I} - (\mathbf{F} - \mathbf{LH})] = 0. \quad (2.43)$$

We choose $\mathbf{L}$ so that $\mathbf{F} - \mathbf{LH}$ has stable and reasonably fast eigenvalues, so $\tilde{\mathbf{x}}$ decays to zero, independent of the control $u(t)$ and the initial conditions. This means that $\hat{\mathbf{x}}(t)$ will converge to $\mathbf{x}(t)$.

Errors in the model of the plant ($\mathbf{F}$, $\mathbf{G}$, $\mathbf{H}$) cause additional errors to the state estimate from those predicted by Eq. (2.42). However, $\mathbf{L}$ can typically be chosen so that the error is kept acceptably small. It is important to emphasize that the nature of the plant and the estimator are quite different. The plant is a physical system such as a chemical process or servomechanism whereas the estimator is usually an electronic unit computing the estimated state according to Eq. (2.40).

The selection of $\mathbf{L}$ is approached in exactly the same fashion as $\mathbf{K}$ is selected in the control-law design. If we specify the desired location of the estimator error poles as

$$s_i = \beta_1, \beta_2, \dots, \beta_n,$$

then the desired estimator characteristic equation is

$$\alpha_e(s) \triangleq (s - \beta_1)(s - \beta_2) \cdots (s - \beta_n). \quad (2.44)$$

We can solve for $\mathbf{L}$ by comparing coefficients in Eq. (2.43) and Eq. (2.44).

As in the control case, this is almost never done by hand. Rather, the functions `acker.m` and `place.m` are used, but with a slight twist. The transpose of Eq. (2.43) is

$$\det[s\mathbf{I} - (\mathbf{F}^T - \mathbf{H}^T\mathbf{L}^T)] = 0, \quad (2.45)$$

and we now see that this is identical in form to Eq. (2.37) where $\mathbf{K}$ and $\mathbf{L}^T$ play the same role. Therefore, we compute $\mathbf{L}$ to achieve estimator poles at the desired location, $\mathbf{p}$, by typing in MATLAB

$$\mathbf{L} = \text{acker}(\mathbf{F}', \mathbf{H}', \mathbf{p})' \quad \text{or} \quad \mathbf{L} = \text{place}(\mathbf{F}', \mathbf{H}', \mathbf{p})'$$

where $\mathbf{F}^T$ is indicated in MATLAB as $\mathbf{F}'$, etc.

observability

There will be a unique solution for $\mathbf{L}$ for a SISO system provided that the system is **observable**. Roughly speaking, observability refers to our ability to deduce information about all the modes of the system by monitoring only the sensed outputs. Unobservability results when some mode or subsystem has no effect on the output.

The selection of the estimator poles that determine $\mathbf{L}$ are generally chosen to be a factor of 2 to 6 faster than the controller poles. This ensures a faster decay of the estimator errors compared with the desired dynamics, thus causing the controller poles to dominate the total system response. If sensor noise is particularly large, it sometimes makes sense for the estimator poles to be slower than two times the controller poles, which would yield a system with lower bandwidth and more noise smoothing. On the other hand, the penalty in making the estimator poles too fast is that the system becomes more noise sensitive.

optimal estimation

The tradeoff between fast and slow estimator roots can also be made using results from **optimal estimation theory**. First, let's consider that there is a random input affecting the plant, called **process noise**, w , that enters Eq. (2.1) as

$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} + \mathbf{G}u + \mathbf{G}_1 w, \quad (2.46)$$

and a random **sensor noise**, v entering Eq. (2.1) as

$$y = \mathbf{H}\mathbf{x} + v. \quad (2.47)$$

The estimator error equation with these additional inputs is

$$\dot{\tilde{\mathbf{x}}} = (\mathbf{F} - \mathbf{L}\mathbf{H})\tilde{\mathbf{x}} + \mathbf{G}_1 w - \mathbf{L}v. \quad (2.48)$$

In Eq. (2.48) the sensor noise is multiplied by $\mathbf{L}$ and the process noise is not. If $\mathbf{L}$ is very small, then the effect of sensor noise is removed but the estimator's dynamic response will be "slow", so the error will not reject effects of w very well. The state of a low-gain estimator will not track uncertain plant inputs very well or plants with modeling errors. On the other hand, if $\mathbf{L}$ is "large", then the estimator response will be fast and the disturbance or process noise will be rejected, but the sensor noise, multiplied by $\mathbf{L}$, results in large errors. Clearly, a balance between these two effects is required.

It turns out that the optimal solution to this balance can be found as a function of the process noise intensity, R_w , and the sensor noise intensity, R_v , both of which are scalars for the SISO case under consideration. Since the only quantity affecting the result is the ratio R_w/R_v , it makes sense to let $R_v = 1$ and vary R_w only. An important advantage of using the optimal solution is that only one parameter, R_w , needs to be varied by the designer rather than picking n estimator poles for an n^{th} -order system. The solution is calculated by MATLAB as

$$\mathbf{L} = \text{kalman}(\text{sys}, R_w, R_v).$$

2.6.3 Compensation: Combined Control and Estimation

We now put all this together, ignoring for the time being the effect of a command input, r . If we take the control law (Eq. 2.35), combine it with the estimator (Eq. 2.40), and implement the control law using the estimated state elements, the design is complete and the equations describing the result are

$$\begin{aligned}\dot{\hat{\mathbf{x}}} &= (\mathbf{F} - \mathbf{GK} - \mathbf{LH})\hat{\mathbf{x}} + \mathbf{L}y, \\ u &= -\mathbf{K}\hat{\mathbf{x}}.\end{aligned}\quad (2.49)$$

These equations describe what we previously called compensation; that is, the control, u , is calculated given the measured output, y . Figure 2.19 shows schematically how the pieces fit together. The roots of this new closed-loop system can be shown to consist of the chosen roots of the controller plus the chosen roots of the estimator that have been designed in separate procedures in Sections 2.6.1 and 2.6.2. The poles and zeros of the compensator alone could be obtained by examining the system described by Eq. (2.49); however, that step need not be carried out unless the designer is curious how the compensation from this approach compares with compensation obtained using a transform based design method.

2.6.4 Reference Input

One obvious way to introduce a command input is to subtract y from r in exactly the same way it has been done for the transform design methods discussed previously. This scheme is shown schematically in Fig. 2.20(b). Using this approach, a step command in r enters directly into the estimator, thus causing an estimation error that decays with the estimator dynamic characteristics in addition to the response corresponding to the control poles.

An alternative approach consists of entering the command r directly into the plant and estimator in an identical fashion as shown in Fig. 2.20(a). Since the

Figure 2.19
Estimator and controller
mechanization

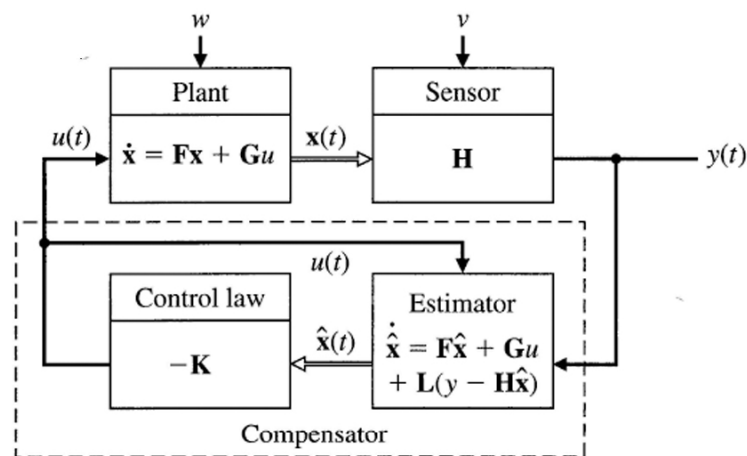
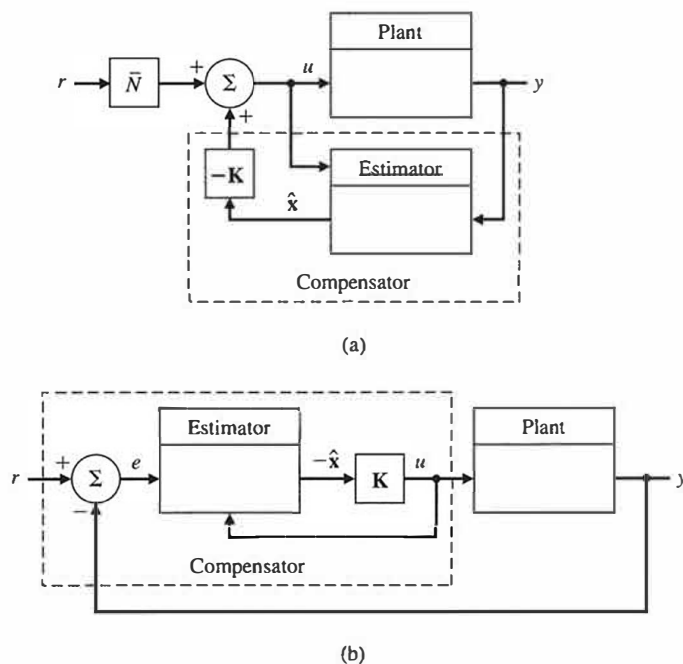


Figure 2.20

Possible locations for introducing the reference input:
 (a) compensation in the feedback path,
 (b) compensation in the feedforward path



command creates a step in u that affects the plant and estimator in an identical fashion, both respond identically, and no estimator error is induced. Therefore, there are no estimator error characteristics in the response and the total response consists of controller characteristics only. This approach is usually superior.

The feedforward gain, $\bar{N}$, can be computed so that no steady-state error exists. Its value is based on computing the steady-state value of the control, u_{ss} , and the steady-state values of the state, $\mathbf{x}_{ss}$, that result in no steady-state error, e . The result is

$$\bar{N} = N_u + \mathbf{K}N_x \quad (2.50)$$

where

$$\begin{bmatrix} N_x \\ N_u \end{bmatrix} = \begin{bmatrix} \mathbf{F} & \mathbf{G} \\ \mathbf{H} & J \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}.$$

2.6.5 Integral Control

In many cases, it is difficult to obtain an accurate value for the plant gain, in part because plants are typically nonlinear and the plant model is a linearization at a particular point. Therefore, the value of $\bar{N}$ will not be accurate and steady-state errors will result even though the model is sufficiently accurate for good

feedback control design. The solution is to incorporate an integral control term in the feedback similar to the integral control discussed in Section 2.2.3.

Integral control is accomplished using state-space design by augmenting the state vector with the desired integral x_I . It obeys the differential equation

$$\dot{x}_l = Hx - r \quad (= e).$$

Thus

$$x_I = \int e \, dt.$$

This equation is augmented to the state equations (Eq. 2.1) and they become

$$\begin{bmatrix} \dot{x}_I \\ \dot{\mathbf{x}} \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{H} \\ 0 & \mathbf{F} \end{bmatrix} \begin{bmatrix} x_I \\ \mathbf{x} \end{bmatrix} + \begin{bmatrix} 0 \\ \mathbf{G} \end{bmatrix} u - \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix} r. \quad (2.51)$$

The feedback law is

$$u = -[K_1 \quad \mathbf{K}_0] \begin{bmatrix} x_I \\ \mathbf{x} \end{bmatrix},$$

or simply

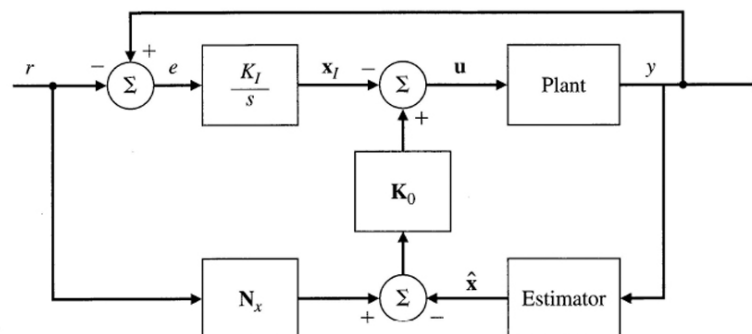
$$u = -\mathbf{K} \begin{bmatrix} x_f \\ \mathbf{x} \end{bmatrix},$$

With this revised definition of the system, the design techniques from Section 2.6.1 can be applied in a similar fashion. The elements of $\mathbf{K}$ obtained are implemented as shown in Fig. 2.21.

2.7 Summary

- System dynamics can be represented by a state-space description, Eq. (2.1), or by a transfer function, Eqs. (2.6) or (2.7).

Figure 2.21
Integral control structure



- The key property of the Laplace transform that allows solution of differential equations is Eq. (2.5)

$$\mathcal{L}\{\dot{f}(t)\} = sF(s).$$

- A system's output can be determined by the inverse Laplace transform for very simple cases or, more often the case, by numerical methods such as `impulse.m`, `step.m`, or `lsim.m` in MATLAB.
- If a system's output is described by $X(s)$ and is stable, the Final Value Theorem states that

$$\lim_{t \rightarrow \infty} x(t) = x_{ss} = \lim_{s \rightarrow 0} sX(s).$$

- One can associate certain time response behavior with pole locations in the s -plane as summarized in Fig. 2.5.
- Control system specifications are usually defined in terms of the rise time t_r , settling time t_s , and overshoot M_p which are defined by Eqs. (2.16)–(2.18).
- For an open loop system given by $D(s)G(s)$, the closed loop system as defined by Fig. 2.8 is given by Eq. (2.19)

$$\frac{Y(s)}{R(s)} = \frac{D(s)G(s)}{1 + D(s)G(s)} = T(s).$$

- The basic types of feedback are proportional, integral, and derivative, and are defined by Eqs. (2.22)–(2.24).
- The root locus is a method to sketch the location of the closed-loop roots of a system vs. some parameter of interest, usually the feedback gain. It is based on phase considerations which can easily be determined graphically by hand, and are therefore very useful in checking computer based results.
- The frequency response of the open-loop transfer function of a system can be easily analyzed to determine the stability of the related closed-loop system. The open-loop transfer function can be determined experimentally or analytically.
- Design of control systems using the state space approach is carried out by specifying the desired closed-loop root location, called pole-placement, or by selecting weighting matrices in a cost function, called optimal or LQR control. Either method tends to reduce the design iterations required over root locus or frequency response design, especially for higher order systems and those with multiple inputs and/or outputs.
- State space design requires that all elements of the state vector are available for the control; therefore, they must be measured directly or estimated using measurements of a portion of the state vector. Pole placement or optimal methods can also be used to arrive at the best estimator for this purpose.

2.8 Problems

- 2.1 Design feedback with lead compensation for the open-loop system

$$G(s) = \frac{10}{s^2}.$$

The rise time should be 1 sec or less and the overshoot should be less than 10%.

- 2.2 Design feedback with lead compensation for the open-loop system

$$G(s) = \frac{5}{s^2}.$$

The bandwidth should be faster than 1 rad/sec and the phase margin should be better than 50°.

- 2.3 For the open-loop system

$$G(s) = \frac{2}{s^2},$$

- (a) design feedback assuming you have access to all the state elements. Ensure that there are closed-loop system poles at $s = -3 \pm 3j$.
- (b) Design an estimator for the system so that it has poles at $s = -6 \pm 6j$.
- (c) Find the transfer function of the complete controller consisting of the control from part (a) and the estimator from part (b).

- 2.4 For the open-loop system

$$G(s) = \frac{1}{s(s+4)},$$

- (a) design feedback assuming you have access to all the state elements. Ensure that there are closed-loop system poles that provide a natural frequency of $\omega_n = 3$ rad/sec with $\zeta = 0.5$.
- (b) Design an estimator for the system so that it has poles that provide a natural frequency of $\omega_n = 6$ rad/sec with $\zeta = 0.5$.
- (c) Find the transfer function of the complete controller consisting of the control from part (a) and the estimator from part (b).

- 2.5 Can you stabilize the system

$$G(s) = \frac{1}{s^2(s^2 + 25)}$$

with a single lead compensation? If you can, do it. If you can't, show why not.

- 2.6 For the open-loop system

$$G(s) = \frac{1}{s^2(s^2 + 25)},$$

- (a) design feedback assuming you have access to all the state elements. Place the closed-loop system poles at $s = -1 \pm 1j$, $-0.5 \pm 5j$.
- (b) Design an estimator for the system so that it has poles at $s = -2 \pm 2j$, $-2 \pm 8j$.
- (c) Find the transfer function of the complete controller consisting of the control from part (a) and the estimator from part (b).

- 2.7 Consider a pendulum with control torque T_c and disturbance torque T_d whose differential equation is

$$\ddot{\theta} + 4\theta = T_c + T_d.$$

Assume there is a potentiometer at the pin that measures the output angle θ , that is, $y = \theta$.

- Design a lead compensation using a root locus that provides for an $M_p < 10\%$ and a rise time, $t_r < 1$ sec.
 - Add an integral term to your controller so that there is no steady-state error in the presence of a constant disturbance, T_d , and modify the compensation so that the specifications are still met.
- 2.8 Consider a pendulum with control torque T_c and disturbance torque T_d whose differential equation is

$$\ddot{\theta} + 4\theta = T_c + T_d.$$

Assume there is a potentiometer at the pin that measures the output angle θ , that is, $y = \theta$.

- Design a lead compensation using frequency response that provides for a PM $> 50^\circ$ and a bandwidth, $\omega_{BW} > 1$ rad/sec.
 - Add an integral term to your controller so that there is no steady-state error in the presence of a constant disturbance, T_d , and modify the compensation so that the specifications are still met.
- 2.9 Consider a pendulum with control torque T_c and disturbance torque T_d whose differential equation is

$$\ddot{\theta} + 4\theta = T_c + T_d.$$

Assume there is a potentiometer at the pin that measures the output angle θ , that is, $y = \theta$.

- Taking the state vector to be

$$\begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix},$$

write the system equations in state form. Give values for the matrices $\mathbf{F}$, $\mathbf{G}$, $\mathbf{H}$.

- Show, using state-variable methods, that the characteristic equation of the model is $s^2 + 4 = 0$.
- Write the estimator equations for

$$\begin{bmatrix} \hat{\theta} \\ \hat{\dot{\theta}} \end{bmatrix}.$$

Pick estimator gains $[L_1, L_2]^T$ to place both roots of the estimator-error characteristic equation at $s = -10$.

- Using state feedback of the estimated state variables θ and $\dot{\theta}$, derive a control law to place the closed-loop control poles at $s = -2 \pm 2j$.
- Draw a block diagram of the system, that is, estimator, plant, and control law.
- Demonstrate the performance of the system by plotting the step response to a reference command on (i) θ , and (ii) T_d .

- (g) Design a controller with an integral term and demonstrate its performance to the step inputs as in (f).

2.10 For the open-loop system

$$G(s) = \frac{3}{s^2 + 2s - 3},$$

determine

- (a) the final value to a unit step input.
(b) Answer (a) for the case where

$$G(s) = \frac{3}{s^2 + 2s + 3}.$$

2.11 For the open-loop system

$$G(s) = \frac{3}{s^2 + 2s - 3},$$

assume there is a feedback with a proportional gain, K , and sketch a locus of the closed-loop roots vs. K . What is the minimum value of K to achieve a stable system?

2.12 For the open-loop system

$$G(s) = \frac{1}{s^2(s^2 + 2s + 100)},$$

use a single lead compensation in the feedback to achieve as fast a response as possible, keeping the damping of the resonant mode better than $\zeta = 0.05$.

2.13 Sketch the locus of roots vs. the parameter b for

$$s^2 + bs + b + 1 = 0.$$

2.14 Sketch the root locus with respect to K for the open-loop system

$$G(s) = \frac{K(s + 3)}{s(s + 2)(s + 1)^2}.$$

After completing the hand sketch, verify your result using MATLAB.

2.15 Sketch the root locus with respect to K for the open-loop system

$$G(s) = \frac{K(s + 2)}{s^4}.$$

After completing the hand sketch, verify your result using MATLAB.

2.16 Sketch the root locus with respect to K for the open-loop system

$$G(s) = \frac{K(s + 2)}{s^4}.$$

After completing the hand sketch, verify your result using MATLAB.

2.17 Sketch the root locus with respect to K for the open-loop system

$$G(s) = \frac{K(s + 1)}{s(s + 2)(s^2 + 25)}.$$

After completing the hand sketch, verify your result using MATLAB.

2.18 Sketch a Bode plot for the open-loop system

$$G(s) = \frac{(s + 0.1)}{s(s + 1)(s^2 + 2s + 100)}.$$

After completing the hand sketch, verify your result using MATLAB. With unity feedback, would the system be stable?

2.19 Sketch a Bode plot for the open-loop system

$$G(s) = \frac{100(s + 1)}{s^2(s + 10)}.$$

After completing the hand sketch, verify your result using MATLAB. With unity feedback, would the system be stable? What is the PM?

2.20 Sketch a Bode plot for the open-loop system

$$G(s) = \frac{5000(s + 1)}{s^2(s + 10)(s + 50)}.$$

After completing the hand sketch, verify your result using MATLAB. With unity feedback, would the system be stable? If not, how would you stabilize it?

